

Converting ORCA σ_{iso} and $\Delta\sigma$ into PAS eigenvalues

A verified prescription for the shielding (Zeeman) tensor, with numerical checks against 1,195 computed nuclei

1. What the two ORCA numbers are

The ORCA workbook reports two scalars per nucleus: an isotropic shielding σ_{iso} and an anisotropy $\Delta\sigma$. The principal-axis-system (PAS) tensor has **three** eigenvalues, so two numbers cannot determine it uniquely — one further invariant, the asymmetry η , is required. This document gives the exact conversion when η is available, the axially-symmetric approximation when it is not, and a quantitative bound on the error that approximation incurs.

The isotropic value is the tensor trace:

$$\sigma_{\text{iso}} = \frac{1}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33}) = \frac{1}{3} \text{Tr } \sigma$$

In the Haeberlen–Mehring convention the principal components are ordered by their deviation from the isotropic value:

$$|\sigma_{\text{ZZ}} - \sigma_{\text{iso}}| \geq |\sigma_{\text{XX}} - \sigma_{\text{iso}}| \geq |\sigma_{\text{YY}} - \sigma_{\text{iso}}|$$

and the anisotropy and asymmetry are defined as

$$\delta = \sigma_{\text{ZZ}} - \sigma_{\text{iso}}, \quad \Delta\sigma = \frac{3}{2}\delta, \quad \eta = \frac{\sigma_{\text{YY}} - \sigma_{\text{XX}}}{\delta}$$

where δ is the *reduced* anisotropy and $\Delta\sigma$ the conventional anisotropy. Note $\eta \in [0,1]$ by construction.

2. Which convention does the ORCA column use?

This was determined empirically rather than assumed. Two observations fix it:

(a) The ORCA anisotropy column takes **negative values** — 20 of 1,066 entries, all of them ^{13}C nuclei with near-zero isotropic shifts (e.g. Tecovirimat idx 9: $\sigma_{\text{iso}} = 0.39$, $\Delta\sigma = -139.94$ ppm). The span $\Omega = \sigma_{33} - \sigma_{11}$ and the quantity $\sigma_{33} - (\sigma_{11} + \sigma_{22})/2$ are both non-negative by construction, so neither can be the reported quantity. A signed value is the signature of the Haeberlen $\Delta\sigma$, which is negative whenever $\sigma_{\text{ZZ}} < \sigma_{\text{iso}}$ (an oblate tensor).

(b) Comparing magnitudes against an independent PySCF calculation on the six molecules whose geometries agree closely (heavy-atom RMSD < 1.5 Å, so the same physical tensors are being compared) gives **r = 0.968** and MAE 6.7 ppm for $|\Delta\sigma|$ across 252 mapped nuclei — consistent agreement on the same quantity, with the residual scatter attributable to method and geometry differences.

Conclusion: the ORCA column is the signed Haeberlen $\Delta\sigma = (3/2)\delta$.

3. The prescription

Step 1. Convert the reported anisotropy to the reduced anisotropy:

$$\delta = \frac{2}{3}\Delta\sigma$$

Step 2. Invert the definitions to obtain the three eigenvalues:

$$\sigma_{ZZ} = \sigma_{iso} + \delta, \quad \sigma_{XX} = \sigma_{iso} - \frac{\delta(1+\eta)}{2}, \quad \sigma_{YY} = \sigma_{iso} - \frac{\delta(1-\eta)}{2}$$

Step 3 (if η is unavailable). Assume axial symmetry, $\eta = 0$:

$$\eta = 0 \Rightarrow \sigma_{ZZ} = \sigma_{iso} + \frac{2}{3}\Delta\sigma, \quad \sigma_{XX} = \sigma_{YY} = \sigma_{iso} - \frac{1}{3}\Delta\sigma$$

The eigenvalues returned are in the Haebleren labelling (XX, YY, ZZ). To convert to the ascending IUPAC labelling used for span and skew, simply sort them:

$$\Omega = \sigma_{33} - \sigma_{11}, \quad \kappa = \frac{3(\sigma_{22} - \sigma_{iso})}{\Omega}$$

4. Numerical verification

The prescription was tested against all **1,195 nuclei** of the 22-molecule set, for which full eigenvalue triples were computed independently (GIAO-B3LYP). For each nucleus the invariants (σ_{iso} , $\Delta\sigma$, η) were derived from the reference eigenvalues, then Steps 1–2 were used to reconstruct the eigenvalues and the result compared with the reference.

Test	n	Result
Round trip with η supplied (Steps 1–2)	1195	max error 1.1×10^{-13} ppm — exact to machine precision
Axial assumption $\eta = 0$ (Step 3)	1195	mean 11.9, median 3.0, max 58.0 ppm
Self-consistency of reconstruction	252	σ_{iso} recovered to 6×10^{-14} , $\Delta\sigma$ to 9×10^{-14} ppm

The error incurred by assuming $\eta = 0$ is not empirical — it has a closed form, which the computation confirms to 8.5×10^{-14} ppm across all 1,195 nuclei:

$$\max_i |\sigma_i^{(\eta=0)} - \sigma_i^{true}| = \frac{1}{2} |\delta| \eta$$

This matters for the present dataset. The asymmetry is rarely negligible: median $\eta = 0.62$, and 84% of nuclei have $\eta > 0.3$. Binned by asymmetry, the mean eigenvalue error from the axial assumption is 2.0 ppm for $\eta < 0.1$ but 18.6 ppm for $\eta > 0.8$. The axial approximation should therefore be treated as a fallback, not a default.

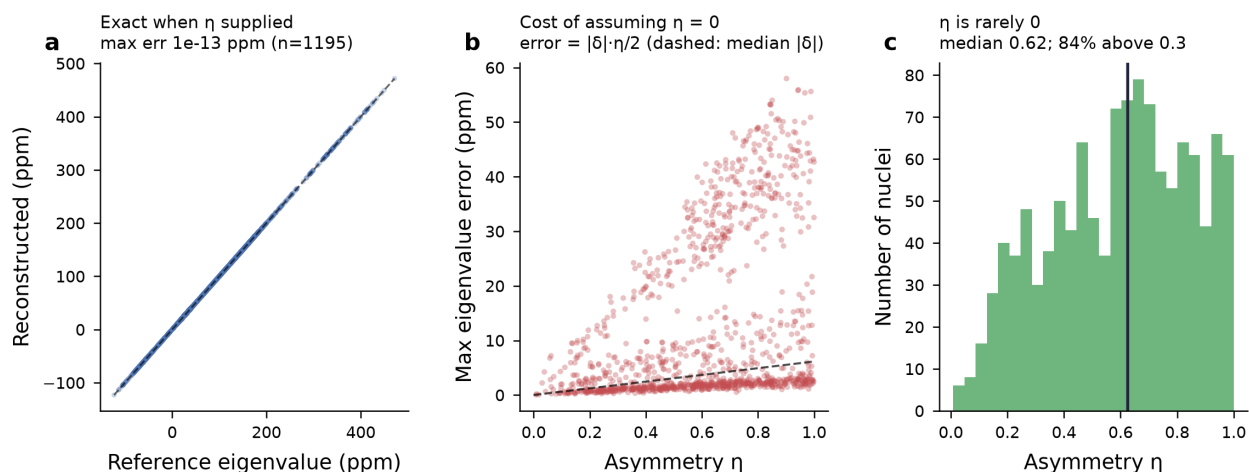


Figure 1. (a) Reconstructed vs reference eigenvalues with η supplied — exact ($n = 1,195$; 3 eigenvalues each). (b) Eigenvalue error from assuming $\eta = 0$, plotted against η ; the dashed line is the analytic bound $|\delta|\eta/2$ evaluated at the median $|\delta|$. (c) Distribution of η over all nuclei; the vertical line marks the median.

5. Worked examples from the ORCA dataset

Three nuclei of Lorlatinib, taking σ_{iso} and $\Delta\sigma$ directly from the ORCA sheet. η is not in that sheet, so both routes are shown: the axial fallback, and the exact result using η transferred from the atom-mapped PySCF calculation. Every reconstruction reproduces the two ORCA inputs to 10^{-14} ppm.

Nucleus	σ_{iso}	$\Delta\sigma$	η	$\sigma_{\text{XX}}, \sigma_{\text{YY}}, \sigma_{\text{ZZ}} (\eta = 0)$	$\sigma_{\text{XX}}, \sigma_{\text{YY}}, \sigma_{\text{ZZ}} (\text{exact})$
^1H	30.525	8.476	0.2341	27.700, 27.700, 36.176	27.038, 28.361, 36.176
^{13}C	158.401	29.369	0.5059	148.611, 148.611, 177.980	143.659, 153.564, 177.980
^{19}F	284.220	114.342	0.6331	246.106, 246.106, 360.448	221.974, 270.238, 360.448

Note how much the ^{19}F case moves: the axial assumption puts $\sigma_{\text{XX}} = \sigma_{\text{YY}} = 246.1$ ppm, whereas the true components are 222.0 and 270.2 ppm — a 24 ppm shift in each, entirely from $\eta = 0.63$.

6. Reference implementation

```
def pas_from_invariants(sigma_iso, Delta_sigma, eta=0.0):
    """PAS eigenvalues (sXX, sYY, sZZ) from ORCA sigma_iso and Delta_sigma.
    eta defaults to 0 (axial). Haeberlen convention."""
    delta = 2.0 * Delta_sigma / 3.0      # reduced anisotropy
    eta = abs(eta)                        # eta is defined on [0, 1]
    sZZ = sigma_iso + delta
    sXX = sigma_iso - delta * (1.0 + eta) / 2.0
    sYY = sigma_iso - delta * (1.0 - eta) / 2.0
    return sXX, sYY, sZZ
```

7. Caveats

- **Two numbers do not determine three eigenvalues.** Without η the PAS is recoverable only under the axial assumption, whose error is $|\delta|\eta/2$ — up to 58 ppm in this dataset.
- **Eigenvalues are not the full tensor.** The PAS fixes the three principal values but not the orientation; three Euler angles are needed for the molecular-frame tensor ($3 + 3 = 6$, the count of a symmetric 3×3 tensor). Neither the ORCA workbook nor the PySCF dataset stores eigenvectors, so orientation-dependent quantities require recomputation.
- **Transferring η between datasets is an approximation.** The exact column above borrows η from an independent calculation on a different optimized geometry. It is reliable only where the two geometries agree (heavy-atom RMSD < 1.5 Å); for the flexible molecules the conformers differ and η should not be transferred.
- **Sign convention.** The asymmetry column of the accompanying PySCF dataset carries correct magnitudes but a flipped sign (all 1,195 entries negative). Use `abs(eta)`; the code above does so automatically.